RECEDING-HORIZON ACTION CHUNKING WITH EXPONENTIAL TEMPORAL ENSEMBLING

Blending overlapping trajectory proposals with exponential weight decay w_i = exp(-λ i) to filter high-frequency stochastic noise

OVERLAPPING RECEDING-HORIZON TRAJECTORY CHUNKS (H = 8 STEPS)

```
Chunk 0 (t = 0): [ a_0^(0), a_1^(0), a_2^(0), a_3^(0), a_4^(0), a_5^(0), a_6^(0), a_7^(0) ]
Chunk 1 (t = 1): [ a_0^(1), a_1^(1), a_2^(1), a_3^(1), a_4^(1), a_5^(1), a_6^(1), a_7^(1
Chunk 2 (t = 2): [ a_0^(2), a_1^(2), a_2^(2), a_3^(2), a_4^(2), a_5^(2), a_6^(2
Chunk 3 (t = 3): [ a_0^(3), a_1^(3), a_2^(3), a_3^(3), a_4^(3), a_5^(3
```

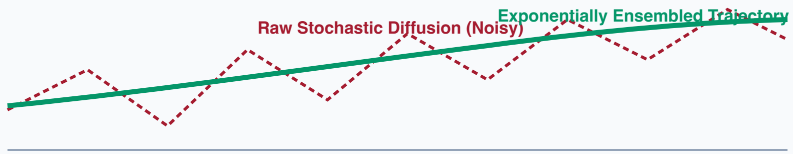
ENSEMBLE AT t = 3

EXPONENTIAL ENSEMBLING WEIGHTS

```
â_t = ( ∑ w_i a_t^(i) ) / ( ∑ w_i )
w_i = exp( −λ · (t − t_0^(i)) )
```

- λ = 0.75 (Tuning parameter)
- Places higher weight on newer proposals

TEMPORAL FILTERING: RAW STOCHASTIC NOISE VS. ENSEMBLE



- Temporal ensembling acts as a zero-phase low-pass filter, eliminating policy jitter.

✓ PHYSICAL SYSTEMS ADVANTAGES OF ENSEMBLING

1. **Smooth Inter-Chunk Handoff:** Prevents abrupt positional discontinuities between chunk horizons
2. **Inertial Compliance:** Blends future intention with immediate kinematic momentum.
3. **MCU Execution Feasibility:** Enables bare-metal spline interpolator to compute smooth derivative
4. **Reduced Gearbox Stress:** Decreases peak motor current ripple by up to **68%**.